

7 (a) Define magnetic flux density.

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..... [2]

(b) A particle of mass m and charge $+Q$ moves at speed v into a region where there is a uniform magnetic field, as shown in Fig. 7.1.

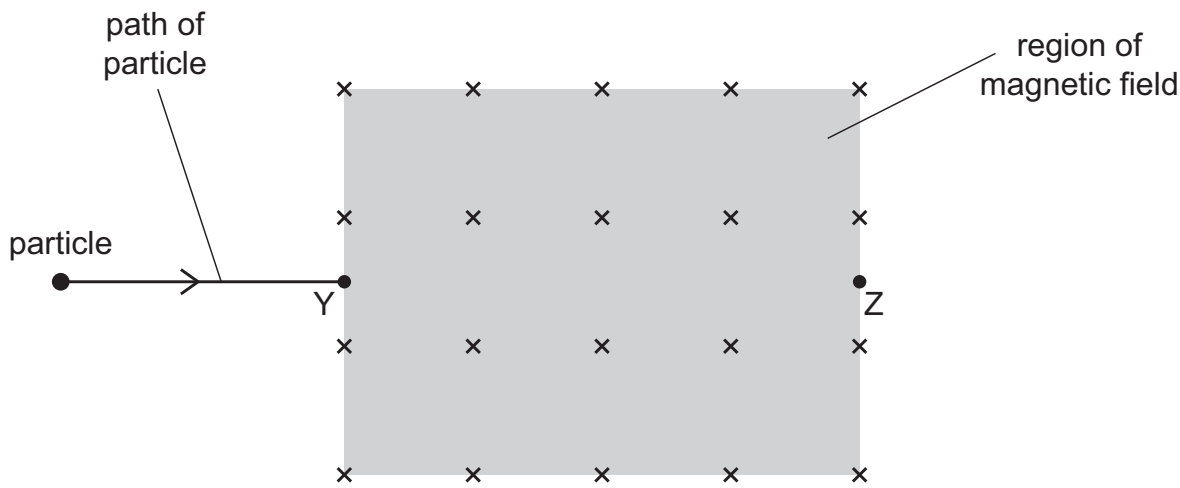


Fig. 7.1

The uniform magnetic field is into the page and has flux density B . The particle enters the region of the field at point Y.

(i) State an expression, in terms of some or all of m , Q , B and v , for the magnetic force F that acts on the particle when it is at point Y.

$F =$ [1]

- (ii) On Fig. 7.1, draw an arrow at point Y to indicate the direction of the force in (b)(i). [1]
- (iii) On Fig. 7.1, draw a line to show a possible path for the particle through the region of the magnetic field. [1]
- (c) (i) Explain how an electric field can be used with the magnetic field to ensure that the particle in (b) now passes through point Z.

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..... [3]

(ii) Derive an expression for v in terms of B and the electric field strength E .

$v =$ [2]